

my name is Alex and today I'll be talking about a work moving towards meaningful pre models for biology if I would to describe in just one word the defining property of the past two decades in biology it would really be scaled we have just gotten so much better at collecting more data faster and to me it's just really astounding how the scaling up has happened in per across many different fields for reporting sequences the number of sequences can in the uniprot database grows annually we are over 300 million sequences in the 202 release in mif the development of robot control microscopes means that we can collect thousand different mly um and just recently the jump CP um initiative releases has released over 100 terabytes of data and the same is obviously true for genomics and omix data the UK B bank now contains 11 pyes of data projected to increase to 40 pyes by 2025 so these big biological data sets are beginning to have a huge impact on artificial intelligence efforts at least historically these data sets will collected prist biology and reference purposes but in recent years advances in deep learning that improved the scale of models meaning that researchers are increasingly repurposing these data set to train massively deeply on your models and so you can see that this is happening across fields which models are emerging across many different domains of biology and Medicine so the vision Behind These large P BS is that they democratize the benefits of deep learning deep learning is an incredibly powerful tool for biology these models can capture patterns and signal and data beyond the capability of conventional methods which greatly improves the biological analysis but the main issue is that they are data hungry meaning meaning that in order to learn these pattern robustly they're going to need a lot of training data and importantly this requirement means that deep leing is accessible to most biologists maybe you're someone who is studying a specialized field and you only have access to a limited number of training examples so the idea behind pre models is to try to forload somean um by by ping the model on a very large quantity of divers data the goal is to initialize the model with a good foundation of knowledge about a modality or in other words ability to recognize Universal Pon that will benefit virtually any analysis so what we do is that we PR this model with a very large quantity of dat of data and after we PR this model we can transer the knowledge learn the biological analysis and in general this is done in two ways the first strategy is called B tuning and it specializes a model with only a small amount of addition additional data so this setup is very useful for example hypothetically let's say that you are a biologist that is interested in producting aspect of protein function but the ass you're using is really expensive and at best maybe you can collect measurements for like 100 sequences typically if you were to go ahead and build a de model off that little data you would not get a general productor but C because the majority of learning has already been done for you you might only need to do a little bit of Tas specific like Nan in order to Ure the general predictor so in other words even if you're work on small B and tuning data set that might be okay one benefit of De about B then is that it allows us to BR tune of smaller data set than be typically feasible without PR the second strategy is called zero short transfer here we're not going to train the model any further and instead we're going to rely upon the premise a pre train model is going to capture General biological patterns and data so what we're going to do is we're going to extract the model's internal representation of data called and embedded these embeddings are a

quantitative series of measurements representing the information that modes extract from input data and how it contextualizes that information and as a quantification of data the the edance of v and can be implied to divers analysis so I really want to make sure that I'm not under selling zero shot transfer in my view Zer short transfer is really fundamental to questions in biology this really brought down to the fact that biology often just involved the unknown not every single biomatics analysis can be reduced to a well defined production problem some times we will not know for certain every phenotype every cell type every function in the universe of a problem and even if you do have some paral knowledge we would not want to Des the discovery process so when you look at the landscape of mtic method out there many methods are really designed to provide exploratory hypothesis about data by contextualizing or visualizing patterns of data not to output for predictions there are many many examples I could point you but among the most famous will probably be bless which has been a main state of btics analyses for decades if you have a protein or genomic sequence that you've newly sequenced you can blast it to retrieve similar sequences in other species and then that's going to lead you to hypothesis about function based upon The annotation present in other species or reasons of um um conservation um um in the in the species so this is fundamentally an information retrieval task right and that is exact the kind of analysis that an embedding from a Zer sh model can do for them if you have a protein sequence you can project it into a high dimension um edance Bas with a protein language model and retrieving similar sequences is as simple as looking for proteins embed or similar but because poent language models project sequences to edden space that is sensitive to structure the advantage to show is that appro that use protein language models like P BL can identify remote homologues of protein that show structure even when the sequences are very Divergent whereas this is more challenging if you were to work with the vanilla um blast firw so in general I hope this explained the vision behind using pain models for biological Discovery by pining on millions of data points we hope to a t model to to General biologically relevant patterns in data and this allows us to these models to project potential nor War data to an embedden space where biological signal is more readily accessible allowing us to visualize compar and contextual data and unprecedented capability so that brings me to the outline for my talk today I spent a good amount of time talking about pron bottles and the potential for EMP power and hypothesis Discovery hopefully I have convinced you of that Vision because the next thing I'm going to talk about is how we actually not currently living up to it I'm going to show you some empirical results across multiple Fields about how preing bottles are actually faing to capture General biological signal when they deplo Z and and I'm going to tr out what I personally believe is a solution who reading meaningful models I'm going to show you some work in meaningful ping task and meaningful inductive biases and I'm going to set the stage for my trainees who will discuss additional strategies and meaningful data and meaningful constraints first to start the SE for the talk let's look at applications in single cell biology these measurements are going to measure gen expression profilis at a single cell level and the resulting data is a matrix where you're going to have a sample of single cells and for each single cell measurement corresponding to the r expression of e visualized asum so a common exploratory analysis analysis with this data is to identify cell types and States single cell experiments often contain

heterogeneous mixture of cells if for example you may take a biopsy from a tissue and then you would expect to have a mixture of different cell types that carry out different functions in that tissue a first question is how many different cell types can be robustly distinguished in the sample and this is an exploratory analysis that is often addressed through visualization so following this really common application space we ask the question if you were to use a single cell model zero shot what would your ability to resolve cell types in a space look like ideally the model should allow us to resolve more cell types than we are able to see in RNA data alone and minimum it should not worse a class of cell types so to address this Kosa who entered with us this year collected a wide range of data sets with well defined cell annotations these data sets are diversified in the tissue type whether cell types of fine grain or coarse grain annotated and whether experiments are collected in single labs of same technologies or decentralized efforts finding multiple labs of multiple different technologies so this reflects multiple possible settings in which experts may want to deploy models to discover cell types recently pre-trained single cell models have exploded with half a dozen being proposed in the past year alone so we took a look at two of the larger and more recent models of public code Geneformer and scGPT which trained on 10 million and 33 million cells respectively to quantify whether cell types are resolvable in the embedded space of these models we use the average bioscore this is the arithmetic mean of three CL metrics the average Spearman normalized Mutual information and adjusted Rand index so basically the higher the score which is bounded between zero and one the better the clustering of samples each of the shapes I'm showing up here is a different data set with the dashed line showing the average performance across all data sets so we compare these pre-trained models against a simple strategy of just selecting the 2,000 most highly variable genes in this data set so you can see that for some data set the model slightly improves our ability to resolve cell types the ppmc um um 12 data set Den noted by the X but for the majority of the data set C actually reduces and drops compared to a very simple baseline data collection strategy and if you were to use a more sophisticated baseline I'm now showing results from SCVI which is an unsupervised variational inference method that is a mainstay of single cell analysis um unlike the previous baseline it consistently beats or matches the SE the strategy using highly variable genes so what does this mean the results indicate that previous single cell models have inconsistent behavior across data sets and unlike established methods there is no guarantee that a model is actually going to improve the ability to cluster cell types over simple data collection I want to convince you that this is not an isolated problem and go to the next point to work by Francis Lee who in fall last year here in San Francisco investigated protein language models which operate on protein amino acid sequences we looked at the alignment of the protein in TASS with domain analysis so to understand this first understand that modern neural networks are hierarchical and they contain many stacked repeats of in CNN blocks as the model gets deeper and deeper it's some embedding of data becomes progressively more nonlinear and complex so to understand this for example think about if you work on images the first few layers may only extract basic information like edges or colors but this information that's reorganized as the model goes deeper and deeper going from edges to shapes to object concepts so typically when you transfer a model you take the embedding from the very end of the model and then you use that for downstream analysis under the assumption

that is the most informative high level concept importance what we did instead is that we Benchmark the performance of analyses using the that each intermediate layer not the start of the final layer the logic behind this is that it could be that the low level features can still benefit the analysis but we don't want just that we want to make sure that the high level concept of proteins that the protein in task is developing um is aligned with a downstream analysis so this open allows us to understand that analysis benefits from the high level features not just the low level features of these models so here is an example of when the P task and the down and the TK are on good alignment the Y AIS is per and the analysis the higher the is the B is the xaxis is um the depth and it goes progressively deeper and deeper blue is a train bottle and owns a randomly in untrained bottle where the dotted line is just the Baseline using the world sequence Lo so you can see that I modu goes deeper and deeper the Blue Line creeps up performance steadily increases but not our analyses behave this way here is an analysis that is not real align here you can see that the task started in performance within the first three or four layers of a model suggesting that it only benefits from the low level feature developed by the task we assessed many times of Downstream analysis and show that most of the analysis actually exhibit this early saturation Behavior the scatter plot shows how much various esm models improving performance in the post blocks that on the xaxis versus the remainder of the model on the y axis each of the colors is a different analysis task with the sizes of circles corresponding to various pent model sizes and you can see that the majority of the analysis task are um at the bottom of the right Axis indicating that they have Zero Performance or close to Zero Performance has that of the first block to recap I showed you across two different domains PR training does not benefit all analyses and data sets while this is going to hurt all applications I think it is especially impactful in exploratory setting because in this setting we may not have a reliable way to measure if models are successful or not so remember in our C clust analysis what we did is that we took some Ro established data sets with ro defin s types and we pretended that we did not know the labels this is how we were able to discover that P single sell models are not all that sensitive to show tapes instead right but in a real world application we simply may not know the underly C in Thea kind of ter them so if a Pon s model misses the soes we may not have any awareness that different method or like a different analysis could do better one consequence of this is that it may just be sa safer for biologist not to use PR mods at all and if so that's not really great because it goes against your vision of use these models making deeply on in Democratic if only biologists who worked on real characterized problems with P to benchmarking can feel confident about using these models that is not a democratic model how do we solve this problem vision is that we must design more meaningful models to explain this let's discuss the pr use to train this models remember that the aim of these models to exp is to expose models on a very large quantity of data this means that there is no guarantee that a preing data points is going to be labeled with any external labels I about function about phenotype or exp annotation what this means is that we have to design a PR in test that only depends upon the data itself and the design of a Pon Tas is actually going to severely influence what it is that the model Laars perhaps the most successful example of pring would be those to tr used to train um L language models which train on corpuses as many as

trillions of tokens work from the Internet pruning is what give these models a deep statistical understanding of language so these L language models use self provide pruned tasks that are designed to help them learn word and sentence semantics and context this is an AO example of a new language models typically use autoregressive tasks but the is called M language modeling and involves masking out sub of Ro shown in gra so that the model can project what these missing words are and hopefully you can see that the logic behind the task is that in order to predict the missing tokens the model needs to be able to understand what the rest of the sentence means the reason I show Mass language modeling is because it is a readily adaptable framework that has already been purposed across biological domains of Protein sequence masks out gene expression values and ask to model to can reconstruct these values given the context of other genes and their identity and similarly protein language models typically mask out amino acids and ask to model to reconstruct these amino acids so there are many advantages to building model of pre-existing Frameworks the biggest Advantage is that we get to tap into best practices and code bases that have been established over use and that greatly accelerates our ability to build models on biological data but to me the biggest disadvantage is that it is not clear what these models are going to learn from this pretraining framework you can certainly reason about them post so for example for protein and language models if you master amino acids in motifs or conserved domains the models can reconstruct these position if it's learned about these conserved elements but is unclear to me if this is universal across our protein sequences what if you're working with a sequence is not real conserved for example so in other words just because a protein task is broadly applicable for natural language is not mean that it is similarly general for biology and we actually have empirical evidence of this remember how I showed you that scg is inconsistent at this point this plot shows you its ability to predict gene expression values as it has been trained to do during pretraining to the mass language model in framework the x-axis is the real expression as C Bend this into 10 sorry 50 equidistant bends the y axis is the model prediction you can see that with the vast majority of genes scgpt cannot form a good prediction and predicts Median expression B you only start to see some predicted performance when you get to the most highly expressed genes at the four weight of a PL in other words the M language modeling framework seems to work very well for housekeeping genes that constitutively expressed across most cells but for other genes at least in this implementation they are not very well predictable and it's unclear if a model is going to learn about the variability or like the conditional expression of this gene overall what this means is that our pretraining methods efforts in biology often take a top down approach they start off by adopting put in task main computer science domains these task learn models are useful for some analysis but not others and therefore what we have to do is that we have to perform a single evaluation across different kinds of analysis to understand where it is that we can apply these models or not this approach is very useful for arguing that models are General but as we discussed evaluating performance requires that you have systematic prior knowledge in places where you don't have this power knowledge and your analysis data set is substantially different from the ones who do have this power knowledge it can be really difficult for us to feel confident about deploying a model instead I advocate for a ground up approach with designing pretraining here what we're going to do is that we're going to start with the biological analysis of interest

Dem mind we're going to think about what kind of information is going to contribute to our understanding of this biology and we invest effort into designing Prest that we feel confident is going to learn this information this um and then we obviously validate that that this signal has been learned robustly um here the P task is still going to be self supervised and therefore still going to be able to scale without labels but the aim is to design more meaningful P where we understand what exactly the protein is on and then therefore we more confident in applying cases where we expect the signal to be useful I'm going to try to make this make this concrete to demonstrate this it guide into an example of where we build BS for the purposes of understanding intrinsically disordered region or IDRs OS are regions the protein that lacks stable secondary or tertiary structure this animation that I have appear shows the structure of a protein the red and the blue parts are the structured regions and you can see that other end of a protein or um um the that are kind of waving around like floppy polypeptide noodles um they don't carry fold into a structure and they all flexible in confirmation this lack of structure IDRs carry out many key functions some of them are involved in protein protein interaction they're flexible so they can adapt the shape to many multiple interaction partners um other those are involved in signaling and regulation they have a lot of exposed um surface area for binding interfaces still other localization base separation and many other function that I have not listed one general characteristic OS is that they evolve rapidly you are looking at the primary amino acid sequence for a um a protein in an ad in the protein C50 acquir 20 different yeast um species the Leu so any with amino acid identity is the same as colored and blue so you can see that while the structured regions Sam and all way that um the adio very highly control um very little of the adio itself is controlled and the conservation is so low that if you were to randomly scramble the sequences the actual conservation is only slightly above that then this lack of conservation at the sequence level means that OS are difficult to analyze remember how I said how introduce BL at of the um with classic methods if you have a protein sequence that you don't know the function of you would source for proteins that have a similar acid sequence and then that lets you make some inferences about its function right that strategy does not work for OS on a global level and as a result OS are fully characterized at a systematic level there's even this review paper that calls them the dark Cross of the dark proteome because even though they are so important we know so little about them OS are actually really abundant in the proteome about 40% of UK carotic residues are disordered this means that on one hand the data requirement for predicting is satisfied even if we were just to predict OS exclusively we would have a very large quantity of data which but on the other hand because of the lack of prior knowledge imper violating that our methods are useful for predicting protein function um is really tough most computational effort focus upon predicting to sequences of the S not what the function of the sequences are the reason B down to a lack of available systematic knowledge with some individual OS we know a lot about the function for most functions we do not know M AIO that carry out the function and for the very few functions that we do data tends to be biased and unrepresentative these labels are often coded through systematic literature reviews or high experiment that focus upon a single organism in a single environmental so hopefully you can see that I why I think IDRs are just a test case like the perfect laboratory for hypothesis Discovery BS given this constraint that we don't have a lot of

knowledge to violate on model with can we still use Pon B to discover hypotheses about AOS here's what our approach is going to be we're going to design a proning Tas that is rooted in biological principle and then that is going to give us control over what p p is going to teach the model in this case we're going to explore principles of evolution and specifically the idea that function is going to diverge lower than sequence as proteins evolve from sh shared common ancestor the the sequences are going to diverge accumulating mutations but these proteins are still going to have to largely retain the molecular functions mutations that drastically disrupt function are going to get selected out so as a consequence you can view Evolution as a natural form of data augmentation it changes sequences without changing the underlying meaning the function at least over reasonable evolutionary distances this principle is going to allow us to corate sets of sequences with evolutionary conserved functions but different sequence identities and importantly we can do this even if we don't know what the function of a poent is we know that recessive sequences likely show function but not always what these functions are going to be for this to be a scalable pon we need a way to create the set of sequences unsupervised and Fort for us there are Decades of established practices and evolutionary pics from for homs that we can tap into here the first principle is that PO that show sequence Identity or likely to be homolog unfortunately we can't do this AIO sequences directly because they devot too rapidly for this principle to be successful at identifying um homas sequences but what we can is you can use the four prot sequences to identify homog sequences and then look for OS within those sequences so to do this we need to calculate an all by all alarm Matrix we need to take every protein in our in a database um calculate how alarm of every other protein and retain the significant matches I'm not going to lie to you this is extremely competitionall expensive but it is also pretty routine operation in btics so fortunately the dat bases that centralize this compation in our work we simply took advantage of oral database which contains his alment for over 20 million proteins so the P previous procedure is going to leave us with infer set of prog proteins and visualize one of these as mulle sequence alarm you can see that the majority of the sequence alarm but there are some Pockets where the alarm is p sometimes not always these are going to be out yours so to convert the set of homog proteins into homog OS we can use that to disorder predictor to predict regions of disordered residues there is one more biological princip we that if a disordered region occurs at the same position ACR proteins they are likely to carry out the same function ACR proteins because the order of the proteins is usually fairly strongly conserved as well so that procedure go explains how we go from homog proteins to sets up unfold homog for our models we want to keep them species specific so we use the spe to independently set PR AIO for used and for human data set the last thing I to explain is how we're going to use the set promos for p we call this P Tas reverse hology at the top these are the sequence for set PR OS we're going to take some of these sequences and then show that to your model in the form of the query set that you see here we'll take out a single hardout sequence from set and you're going to mix it up with 399 from other proteins um random sample from the prot down the model has to transform all of these proteins both in the quer set and the target set into embedd and that's on the basis of these embedd which of the target sequences is a hom from that a previous ment of mine Amy L developed some Theory supporting this

method that you can read about on the paper but to somewhat the main points we think that solving the reverse mology Pon in task is going to require the model to do two things first it's going to need to recognize features of the sequences that are conserved through Evolution the second thing is that it's going to need to BL features that are um that will distinguish aios from other aios that are General and shared across mod most AOS are not going to be able to disting with different homology families from each other we think that both of these properties are going to lead us to interesting hypotheses evolutionary control featur tend to be important to function and the state feature distinguishing feature tend to be more specific to the function of a poon even in the absence of systematic Benchmark data sets there are a few strategies that we can emplo to validate all methods here we asked how many po functions are discoverable fromers hology edance compared to knowledge based features these knowledge based features are features that have been coated for from over 30 years of Audio Research and have been shown in literature to be important to the function of at least some previous Studies have shown that if you were to take this feature and cluster them the cluster own um by function for aios so what we did is that we compared the features feates on the ability of cluster function of an unbiased set of functions um we quantify how Clump and non redundant sety annotation R inside of button space the y axis is both enrichment with rology features the xais is build enrichment with knowledge based features so you can see that from featur the cluster at the bottom um left right um there isn't drawn inment other featur this is probably because that these are function that audio don't really carry outcome although we can't R out that these a function that NE future is able to discover but for the function that we typically see a strong investment for using knowledge Bas features we often do see an investment for using rology features and moreover if you look at um the um left top part of axis you can see that there are several functions that are enrich in using the reverse hology edance but not with knowledge based features so this suggest that not only do own B ReDiscover function that have been coated and um from 30 years of like PRI um knowledge and AIO research but we expand to discover new functions as well at this point we are confident that our model has learned both from Theory and from comparison to leadership now we can finally address the question how do we produce hypothesis in this model because we knew that this was going to be a goal from the very outset we made design decisions along the way that allow interoperability one was in the design of architecture each no in a new network produces the score at each position in the sequence this is not an inherently interoperable architecture but its properties allow us to visualize each of the features bi model as a sequent logo because we canor through all OS collect the subsequences that produce the higher score for and a LOM that of a sequence logo so in other words there trate feature dra by a model from which is a blackb model to a visualization commonly used in btics to understand motifs and patterns and here you see in a summary of all of the features bi model these sequence logos represent globally the kinds of sequence features or model thingss are conserved across OS and just visually there already plenty inform pH analysis so for example here's one feature energies model and resembl the pka with the poin kindness a motif this Motif allows for PKA to boun your sequence leading to the phosphorylation of the cine post trans modification but we also observe that this model developed a second independent feature that kind of look like

two overlapping PKA motives and the reason why this feature really C right is that because at the time of a work um another group independent reported for at least one protein and one PKA is capable of phosphor in two s in a row depending on both serin if phosphorated or if just one of the S is phosphorated the function of the protein is altered So based upon the fact that our model would dedicate a separate independent feature to this overlapping with we hypothesize that this mechanism might be more wi than nthl alone to test this we went to biom bioman which um creates experimentally identified post modification sets we found that in the sequence window that maximally activate the second feature you see nearly three times the number of sequences with multiple adjacent Frost relation States compared to that of like the first fature this suggest that our future may be picking up upon a more widespread pattern hereo and hopefully as I hope this illustrates how interpreting P models can help us discover Global hypothesis about AOS in addition to producing hypothesis about global course OS we can use interpretation method to produce hes about specific independent OS suppose that you have an AIO that you are interested in and you would like to generate hypothesis about what features are important to the AIO you can directly take the audio and analyze how Fe that are most important to your audio behave quantify this by taking the most activated um features the AIO compared to that of the rest of the in this case we looked at um AIO in prot H1 hn rmp A1 which is a face separated umary binding protein that is implicated in Cano I visualized just one of the most important features of the protein um as what we call a mutation Le map them so you can see that the H the map is above the axis the more the amino acid in the position contribut to the Future the most fa um amino acids are shown above the axis and the least fa are shown below the axis so to interpret this you can see that this feature really likes aromatic amage there are a lot of F and Y across above the axis and similarly Le run these aromatic of PL by G's and s's you can see that there's a lot of G's and s's in the regions around the Y and F this result is significant because is actually consistent in what is known in literature about this protein this kind of feature aromatic aets blanked by G's and s's is the feature that promotes weak stack interaction that allows spent to face SE so in this case this model has correctly identified a biologically relevant future for aort we know a bit about them suggesting that it may have the capability to do this for audio that we have less knowledge about suiz I showed you how designing meaningful P task can improve our confidence and applying PR B to more unknown biology the next thing I want to show you you is to show you a different option building meaningful models meaningful inducted biases this is work in part progress unpublished there um story hasn't fully solidified but minimum I do want to show you a teaser of this work as a further example of a design philosophy here for this part we're going to look at an application of microscopy morphological profile here F microscopy images are going to be used to drug screening let's say that you have some cells that reflect some kind of an unhealthy state so for example coron virus infected cells these cells often have changes in morphology that might be too subtle to SP by human eye but can be reliably detected by um computer vision methods to discover drugs we can those sells of compounds and identify the compound that we bu the unhealthy phenotype back to that of a healthy one these screens often involve massive libraries of as many as hundreds of thousands of compounds so cooperational methods own to mental cure for scaling the

analysis of each data databases this is also a multimodal problem right we have the images but we also have information about what chemical compound just sells in the images a lot of biological problems of multimodal not just myological profiling but multi py images where you have clinical notes the rest really goes on so because of this there has been a lot of interest in using um learning multimodal embedden spaces and many of them full inspiration from the methods used to train text to image models like D or St um diffusion where you can type in the caption and then generate an image of that um corresponding to the caption these methods pre-train a draw embedd space so texting images show the same space given the embedding caption images that Master caption should be very close together in that space this p is now similarly emerging as a strategy for multimod biology with some work now proposing learning Jo compound in image space and beyond that for other modology for morphological profol um in the same way however learning a dra Ed space may not make sense depending upon the modalities that you're working with genotype phenotype peration don't really belong at the same level so to see this let's consider the case where we have two images of two different cell types that have intruded of the same compound these SES may have different um phenotypic responses to a drug maybe one type of cancer St is susceptible to a drug but the other is non responsive so now it becomes unclear where it is that we should un the compound we put it by the first by the secondine or somewhere in between in other words a your embed space between assume that compound and images are directly comparable but in practice we don't compare compounds of images as much as we compare the impact of compounds to see this in a morphological profing experiment starting from C culture a borrow just really has two options the can either do the cell of a chemical compound and an image of the cell's response to the compound or they can use a mark treatment to absorb what the cells would look like if they were left unteted we can never really observe both images to the same cells at the same time so we call these images cter fractures of each other but this counterfactual perspective mological profile helps us understand how compounds and images should out to inter a compound should transform images converted and cell to a counter of two the this is the principle that we're going to build into a model as an inducted bias Jam or sum inter created the method that we call Micon and here's our set up we're going to use an image coder shown on the left to embed compound images shown in red and unteted images shown in blue and we're using a mo code top to embed compounds instead of comparing the compound and the image B directly we're going to use the compound and B to transform image B starting with an unteted compound blue we inject Mo B through an added model to produce in synthetic TR and to make sure that we're producing realistic sythetic treatment we compar to re bance the treatment and minimize the difference and finally we can do the exact same thing with the Ted using the mole EDD now to subtract from the image EDD which you can compare to the real butons of unshed images in the exact same way formally larning is Ace Contra of loss I'm not going to go deep into this but this loss essentially allows us to scale a goal of minimizing differences between real and synthetic images at scale by creating many synthetic comp embedd and many real embedd and then comparing them all against each other at once this is a work in progress so I'm just going to show you a proof of concept developed on just 55,000 images now but even with a relatively small data set our model is infective producing informative edance we quantify this by assessing if

um embedd are reproducible or course replicates if you were to those cells with the same compound in two independent sets of experiments how frequently would the nearest Neighbors in the second set of experiments be that of the same compound in that of the first set of of experiments versus the different compound this is Quantified as an aacy score based upon if you retrieve a March versus a nor Marchin compound um so this retrieve task should have a pretty low um upper bound because not all compounds IND induce significant impacts upon f but for the compounds that it that do induce an impa phenotype is a g measure of the Ed a extracts information about reprodu biology and be a robust B effects that do not replicate so here we Benchmark the retrial accuracy for 187 compound seen during training um and 103 compound unseen during training again hand engineered of Prof feature that good performance of what dat test for morphological per for over um decades and in fortunately you can see that we outperformed the so portfolio Baseline suggesting that even in a SM St proof of concept we are Lear in signal but the most exciting part of a method is that it predicting the impact of compounds is built in we can generate synthetic andand of cells to the compounds potentially never see own training and predict what the edance in the compound space might be here's the tasty of the embedding of eight real compounds in dark colors compared to generated embedd of these compound colors and L I want to emphasize that we are not doing perfect you can see that there is a distribution Tri between the real and generated in beddings probably because there are a lot of Sy compa from go to go that are difficult that to get SP on however you can see that at least for some compound the real and generated distribution are closer for that of marching compounds than that of the real ones and so while we know that our gen BS are not perfect they are often reasonably close when we generate compound in in training um and which the nearest real nebes they have a 23.5% accuracy March in the compound this is actually pretty good if you call that the accuracy to retrieve in real images is 32% which serves upper bound um mostly you actually have some performance even when you're working a compound completely unseen in a TR data set about 20% accuracy for 103 unseen inen compounds but the ability to generate image bed gives us a bur rate to assess and when Real Models can generalize this graph is a compound by compound breakdown of the r accuracy for unse compounds and you can see that while some compounds are match perfectly many have no accuracy in the book so one question that we can ask as we continue to develop this work is how is acacy influenced by various factors for example do compound similar to that of in tring structurally similar and to have high to conclude and recap the main points of my talk today that I argued that pre models can greatly accelerate hypothesis Discovery in biology but I noted that the major limitation is that current pre BS are inconsistent preventing The Confident deployment in exploratory settings and I propose that we develop more meaningful model that control what it is that the model launch alar learning Tas and given us validation strategies I showed you two ways of unit through meaningful Pon task and meaningful inducted Biers but these are not the only ways of using PR knowled to design models my trainers are going to show you some strategies using meaningful data and meaningful constraints the impression that I hope you get from all of these Tru together in synthesis is that developing meaningful models model and a spectrum with a no one siiz or strategy so for example I showed you that meaningful ping test can be used if you have some power knowledge about the overall

or general signal that a model should learn but sometimes we are not going to have this but we may know for example what kind of information that a model should not learn and Alex LeCun is going to show you some strategies of imposing these meaningful constraints in other words the strategies help us incorporate prior knowledge in different ways and at different stages of model development and design and multiple strategies can be used at once I'll be talking now about meaningful pre-training data building off of what was talking about a while ago and I'll try to convince you from work that we did two years ago that creating meaningful pre-training data is one way that we can learn how to create models that have features that we might be interested in for our downstream tasks and so I kind of want to motivate this by a little bit of an easy brain teaser so I want you to think about something that's expensive noisy and comes in small quantities and if you were thinking about biomedical imaging data and hopefully my thing would pop up yes if you're thinking about biomedical imaging data you'd be absolutely right and so that makes it very difficult for us to be able to whoa sorry that makes it very difficult for us to be able to train models completely from scratch so that's why as Alex mentioned one cheap alternative is transfer learning and this is where we take models that were trained on a different task and apply it on another data set where label data might be scarce and so we've seen this been useful in many different situations uh we've seen it been used for breast cancer classification a long time ago we've also seen it been used for diagnosis of skin diseases as well as histopathology data now I want to kind of give a brief uh refresher I'm sure many of you have heard about transfer learning before and perhaps even applied it in your own work can I get some head nods to see if uh you've used it very likely right but I'll just give a quick uh brief overview of what it is again so transfer consists of two things right one is where uh we have some data set A and we train a model on it right but we're really interested in applying the model that was trained on some data set A and applying it to some data set B with far fewer labeled examples right and so as I mentioned this has been used for breast cancer classification as well as skin diseases and for the examples that I showed in the previous slide ImageNet was the data set that was used in order to create these pre-trained models and so I'm sure you've also many of you have heard of ImageNet as well but I'll also revisit this so ImageNet is this large scale data set of natural images with very diverse sets of uh a very very diverse set of images where each image could be one of 21,000 labels but whenever someone really talks about ImageNet they're really talking about ImageNet-1K which is a subset of ImageNet with 1.3 million images and 1,000 labels and so this was used in the ImageNet large scale visual recognition challenge beginning in 2010 but it was only in 2012 when the first deep learning model was used on this data set and that showed far better performance compared to all the classical computer vision models at the time and so that's what dominated the field following that and at some point we noticed that our deep learning models perform better than humans so that's why the comp competition was retired but we believe that because these models were able to perform so well on such a diverse data set it was able to learn very diverse set of features which are robust to different situations right and so that's why people believe that good performance is correlated with good transfer good transfer of these features in downstream tasks and so one issue is thinking that simply because it's learned a very diverse set of features

that it'll be applicable to any task right and so what we want to think about was that there are no guarantees that simply applying a model that was pre-trained on image net will work well on any data set like for example microscopy images and so features that might be useful for natural images may not be useful for microscopy images if we think about features that might be shared between them we might think that um a model that's learned about shapes about texture and about all these general purpose features might be useful whereas in natural images specifically we might be more interested in features that capture things such as depth or the relative position between these images between object in the image whereas with microscopy we don't really care that much about that and perhaps we want properties of our features that are different from natural images for example we might want features that are rotation invariant or perhaps uh if uh features that are also magnification invariance so it's that's to say if we did zoom in to an IM a microscopy image perhaps we would want to have extracted features that tell us the exact same thing even if we zoom in or if we rotate it right and so that's not necessarily captured by pre-training on natural images and so now we were thinking instead of simply thinking about is image our image models image that pre-trained models useful for microscopy data sets perhaps we should be thinking what type of data should we be using to pre-train these models right and so that's where we believe we we POS the research question can pre-training on a large diverse and more domain specific data set yield us more useful features for the downstream tasks that we interested in and so at the time there was not as much there was a data set that was as diverse as we had wanted to so the next logical question was can we create a data set for microscopy images which rivals imet in its diversity of the types of images that we see right and so what does that ideal data set look like so that ideal data set would be large in the amount of data we have in the amount of labels that we have as well as being diverse in cellular morphology magnification possible treatments that we use and most of all uh diverse in data sources and so that brings us that brought Alex me and Alan to create cyto imet and so doing an Apple apples comparison cyto imet is a large data set with 890,000 images and 894 classes um and these come from images from 40 publicly available microscopy data source so one thing that I want to point out that is a big differentiator between C imet and image net is that whereas in imet all the labels were humanly uh they were annotated by humans inside of image net we assign weak metadata labels based on whatever met metadata we were able uh to retrieve for these images and so I want so now that brings us the question how was Cyro imet created and so we were running under the hypothesis that a more diverse data set will yield more diverse features and so we thought of diverse uh diversity in three different ways right we thought of diversity in data sources and so we we sought to get uh images that were from very different databases and very different experimental setups right we thought of diversity in terms of the types of microscopy that we use as well as and as I mentioned we use diverse experimental setups so what we end up with is a data set with a great number of organisms that are present many different types of stained proteins cell lines phenotypes genes that were specifically knocked out srna and so on right and so we believe that having such a diverse data set and exposing our model to this will be able to EXT uh create a model that's able to extract features that are generalizable to all these different uh diverse use cases and so what's one big difficulty

with this microscopy Imaging data is is very messy and so a lack of standardization between these data sets makes combining a lot of these diverse data sets a lot more of a difficult task and so we see lack of standardization in many different formats we see a lack of standardization in how we download the data right how the image directories are structured how the multi- channels if if the data set has multiple uh channels that was stain for how was it stored in some cases if they have three channels they combine it and make it like an RGB image whereas in other data sets they might have these microscopy channels as separate images altoe different image formats some being public and some being even proprietary and then the way the metadata is also stored is very different and some don't even have metadata files at all so one unique challenge that I did want to highlight that really shows the difference between natural images and microscopy images is that microscopy data can have any number of channels so whereas in natural images we have three channels one for red green and blue in microscopy images each of these channels stain for distinct in different uh types of proteins right and so we can have microscopy data sets with one however many number of channels as we're interested in right so how do we standardize data with this many different number of channels so in order to do that we opted for a more simple approach and tried collapsing all of these channels into one grayscale image so now we have a data set full of grayscale images and we did that by first normalizing the pixel intensities within each Channel and then we just simply averaged overall of these channels to create one grayscale image um for these multi-Channel images okay so now I kind of want to highlight some strategies that we employed in order to learn diverse features in our pre-training Tas and so these were design decisions that we made on our data set in order to influence our perception of what useful features we might be able to learn Downstream right and so I'll be mentioning this in the context of what are issues or differences between natural images and microscopy images and then like the design choice that we made and so for one example compared to imag net it's very difficult to assign microscopy images only one label on the image on the left is an image of a dog or a boxer dog specifically but on the right there's many different ways that we can describe uh this image right we can say that it's um an image of human cells specifically huvec right we can mention what proteins were stained what was the phenotype that was observed or and what compound was used to treat it right but we're not so much interested in one of these particular things but we're more interested in being able to train a model that's able to generate features that can help us understand multiple aspects of our data without honing in one specific uh part of this metad there right and so in order to do that we designed a weekly supervised pre-training task which predicts one of the metadata labels that are present here right and so for this image we just randomly oh well we chose labels that try to maximize diversity in terms of how many data sources are present right as well as diversity like within class so in this case this image was assigned a label of hvec right whereas maybe another similar image might have been assigned perhaps the phenotype right and so um that's the that's the way that we were initially thinking about how we were going to create just the weekly supervised pre-training test and so one other one other issue that we faced was that our data set is large but severely imbalanced right so one of the DAT like our smallest data set that we collected had less than a thousand images

whereas the largest data set that we collected or that we CH we got had over 150,000 images all right so this imbalance in data source tended to go into how we got into the classes that were assigned right and so if you look for example for images that were labeled hvec I color them based on those with similar metadata let's say and those let's say that for this specific label we see a lot of uh images that come from the same data source that are like say in Orange right so the next thing is how can we promote class diversity if we have such an imbalanced data set so our approach that we downsample the data set stratifying by unique metadata as much as possible and so the result would be having a class with images from as many different data sources in as many different experimental setups as much as possible and so this greatly reduced our initial data set size of 2.7 million images to 400,000 images uh but we believe in the process it created classes that were a lot more diverse and a lot more difficult to tell apart okay so one of the hypothesis that we had for microscopy a useful microscopy features is that they were invariant to the magnification that you chose right and so how do we promote learning of features that are robust to varying magnification and so we did this by artificially increasing the diversity of image resolutions by sampling image patches of different sizes within our data set so if we take an image from any one of these classes let's say we separate it into panels of four quadrants and then we choose a scaling factor we randomly choose a scaling factor in the Power of Two And we apply that to each of these uh crops essentially so after that we normalize these crops uh we we normalize the pixel intensities with respect to these crops and then we choose whether or not to add one or four of these one to four of these patches back to the data set and so by doing so we artificially increase the size of our data set from 400,000 to 894 th000 one thing I want to note and is important to for my last for what I've shown about the pre-training task but what's important is that when we we were creating this data set we weren't interested in using this data set as a benchmarking data set compared to imag net for example we were more so interested in trying to create a data set which with which we can use for some for just a pre-training task right to learn features that would be useful for Downstream microscopy tasks and so now that leads us back to the same issue classes still remain imbalance by Source data set here's a glimpse of what how many images came from a specific database in the existing in the existing cyto imet data set and so you can see a large proportion of our data comes from these recur these large recursion data sets with and then we see that there's a large number that come from other places but still significantly from recursion and the big issue is that that a lot of our labels come from one data set right so over half of them come from one data set so how can we avoid the trivial solution of Simply separating these labels by data set and so initially we tried to explore that by creating U maps of uh these of a subset of our classes to see how well uh these classes were already being separated by data set and so what I what you can see here at the left images or a lot of these like small clusters with images from the exact same label right and so we thought that was problematic but fortunately it looks like as a result of increasing diversity in resolution because of our patch based upsampling we see that a lot of these clusters um start to disappear right and so that can tell us that our data set has become a lot more diverse and a lot less easy to Simply T apart by data set alone right and so that was useful but now another hypothesis that I

wanted to touch upon was similar to our idea of wanting features that were that were robust to or sorry that were invariant to magnification we also wanted to learn features that were robust to rotation and so that's a lot more simple because we can add rotation as an augmentation during our pre-training task okay so this I wanted to touch upon is the details for our pre-training so for cyto image net we took 90% of the data for training and 10% for validation uh for early checkpointing and so a comparison between how models were the model was trained between side of image and image net with these image net pre-trained models a lot of the time they're working with a lot of compute right and so they can afford to be able to train for a lot of epoch with as many gpus as they want right whereas we were operating with only one GPU so we only needed to train this for we ended up only being able to train this for 24 Epoch and that alone already took two weeks right okay so now I want to talk about the way that we evaluated uh these pre-trained models so we wanted to be able to compare features that were extracted from image net pre-train models and cyto image net pre-train models to see whether or not these features were robust to different uh to varying tasks right and so we use three Downstream data sets the BBC bbc 021 data set Cyclops data set and C7 data set and these data sets we chose specifically because they have very different they have very different uh types of data or different types of data that we were interested in being able to see whether or not our features were robust so for example for the first data set U for the first data set we uh it wanted we needed the task was to predict the phenotype based on cells that were treated with a specific compound right and so this requires features that are specific um or that are descriptive of cellular morphology whereas with um the second task which is subcellular localization we might want features that are more robust and insensitive to Cellular morphology right so we test we end up testing our feature on a very diverse set of data sets coming from very different uh organism types right single cell images versus whole scale a whole image a whole images right and we report their K near neighbors accuracy on the features that were extracted alone so one thing I want to drive home here is that all these Downstream data sets were not included in pre-training right neither were they used for fine tuning we simply extracted features for all of the data in these data sets and then use K nearest Neighbors on the test sets of these data sets Okay so now that leads us back to our research question can pre-training on a large diverse and domain specific data set yield us more useful features for Downstream microscopy tasks and what we find is though cyto image net features do learn compare features that are competitive to image net they actually don't outperform image net features and so this came to us a surprise initially but what this told us was that image net features were a lot more uh robust than we thought but at the same time this also meant that our pre-training task did yield this features that were uh useful for these tasks right so I guess one the next question that we had to ask was did pre-training on cyto image net versus image net really change the features that we did learn right did they learn similar or did they learn conflicting features which helped them on these Downstream data sets and what we found was by simply concatenating their features we end up with feature a feature set that's a lot more descriptive than either of them alone right and this this results in improved significantly improved performance and this suggests that perhaps these perhaps these we can augment these general purpose feature extractors the ones that were pre-

trained on imet or these other diverse data sets with more domain specific pre-training in order to achieve better performance on these on whatever Downstream task that we're interested in so that was work that was done two years ago but what has changed in the past two years and what what can this work tell us about more current more recent pre-train models right so in the past two years we've seen a lot of methods that were very successful pre-training on domain specific data right we've seen some that perform weekly supervised pre-training and many many more recently that perform self-supervised pre-training especially on unlabeled data right and so what does our work on cyto imet say right it says that it suggests that perhaps these domain specific pre-train models can be further augmented by in by combining them with these general purpose pre train models and so more recent work has shown that domain specific pre-training beginning from general purpose pre-train features can actually improve domain in domain performance and fine-tuning on out of domain that out of distribution data sets and so one key difference between what they did and what we did is while we simply just concatenated the features that were uh ex that were learned by these two models um the work work by Google they actually performed in domain pre-training self-supervised pre-training after initializing from these um these general purpose features as you can see here and then they do fine tuning after and so this is where I kind of want to leave uh leave or end the talk with and that's that perhaps although we've seen a lot of usefulness in being able to train on these domain specific data sets in recent years perhaps there might be value gained from being able to combine them with a lot of these general purpose feature extractors that we're gaining from other mainstream CS uh data sets so I'm Alex I'm a PhD student at Harvard um I'm going to be talking about some work that um I did with Alex Lou when I was an intern with him um basically on how we try to develop a method to correct for batch effects within deep learning models and um I'm going to kind of pick up where Stan left off and talking about you know how we can meaningfully design um you know training strategies for models to kind of extract the biological signal that we're looking um so I'm going to be focusing on microscopy for micros specifically because that's what we worked on um and as many of you here know I'm sure that deep learning is becoming a very promising tool um for evaluating microscopy images and doing microscopy analysis um and this has been the combination of a few things so first like we have you know incredible ways to generate High throughput um images and uh we can conduct like many experiments in large scale and and collect many images as result from them with Technologies like cell painting and others developed here at the road um and then concurrently um in the Deep learning world there's been a lot of advances in modeling um that have kind of driven uh computer vision tasks for natural images to be um better than human performance like Alex was mentioning and um kind of this kind of begs the question you know can we Port some of these uh models that have been trained to do very well on natural images to also do well on biological images right and I really like this uh review that was um done by members here at the broad about um all the different ways in which deep learning has been used my CR we have like super resolution we have a molecu to image synthesis segmentation Etc and uh one of the most exciting applications in my opinion is morphological profiling um so given like the techniques with cell painting and others we can you know we can create many different types of genetic and chemical perturbations

and then use um microscopy to kind of image all the effects of all these different permutations and um hopefully with deep learning we'll be able to for every image learn some feature Vector that encodes some significant information about the phenotype right that has been observed from the um and this is really exciting because you know assuming that this deep deep neural network can actually extract relevant biological signal um now we basically transform these like peration effects into uh a numerical Vector that we can then analyze right and so you can imagine like comparing like different vectors corresponding different perations and doing analysis this way and this can lead to you know advances in drug Discovery and functional genomics maybe we'll that certain perations right their feature vectors are closer to each other so those drugs maybe they have similar mechanisms of actions or we might find that there are some outliers right so some features that maybe are different from any perturbation that we've ever seen and that could help us discover kind of new things so this is a very active research area and people at the broad will definitely know this especially people in the Imaging platform right there's been many many different methods over the past few years that been uh designed for how do you train the Steep in your own neck right to extract you know these feature vectors from images um and um however there there is kind of a recurring problem that we we often come up and this is something that you know computation biologists will know well which is uh this this problem of batch effects right um and uh you know this is this is like a a common problem like I said like you know you go into lab you collect some images right um like from let's say a particular treatment you collect different another images from another particular treatment right but the problem is that um when you were collecting those those data points um there was a particular like temperature right there was a particular humidity you were using a specific instrument right and then let's say you go back to lab the next day and you want to kind of connect collect the same type of you know data but you might find that you know the temperature has changed humidity has changed and like someone sole your favorite microscope so you have to use like something else right and um you know you you perform the same exact like experiments with the same like treatments and even though you you know keep all of like like let's say the biological principle is the same what you end up finding is that your images look very different right um so there are some like sources of technical variation that have kind of uh been uh incorporated into these images that could bias Downstream analysis we don't find a away to um and to give a very concrete example of how this can manifest right let's say I train like a deep neural network on data I collected uh from batch one right say train a network to predict like the treatment right from the image um and then I want to test my network on images collected from another batch um so this is this is like real data real images and real results from the uh recursion RX RX1 microscopy data sets but what they they find is that you know in training you can kind of achieve 100% accuracy you can classify these um these treatments very well um and if you were to validate the model on an independent uh set of images collected on day one for batch one you get like 40% accur accuracy right so that's like the generalization um ability but then if you were to evaluate on a different batch um then you you see like a significant drop in performance right so there's like there a gap of 10% here that is due to the fact that you know you you have to deal with the fact that these

data points are collected from different batches um and so that's one way that it might manifest right your your accuracies whatever they might not be so uh so so good but another way that batch effects might manifest is remember in morphological profiling I talked about how we're learning like these neural networks that can kind of take an image and produce a feature Vector right and we want to use those feature vectors for Downstream analyses of of of drugs or genetic perations or so on and so forth um but if we were to like let's say take you know a bunch of images and get their respective feature vectors and then like let's say project them down to a few Dimensions that we can kind of visualize so each dot here is like a different image right with a different peration what we find is that in images in the same batch um kind of live very close to each other within this feature space and and they they cluster according to like these these batch identities right which could very well be like a problem for Downstream analysis right there's this this artificial technical like bias that we see in our data and so um so I'm gonna now talk about like a method that Alex and I we worked on last summer um it's called batch effects normalization um it's a specific method that we try to develop for for correcting some of these batch effects in deep coding models um and it's it's a relatively simple method like the the way it works it it draws inspiration from kind of classical normalization techniques that have been used for like many years in computational biology right um so this this idea that you know okay so I have like you know kind of two different data sets they're from different distributions what if I just like do some sort of standardization right so like I can take the mean and the the variance and I can use those to standardize my various data sets to try to align right those those statistics um so that I can get rid of batch effect so um you know again this has been used a lot because it's it's simple it's intuitive um but what what has also been shown um in in various settings is that for deep learning models they don't seem to to work that well um and I think intuitively the reason being that you know these normalization techniques they are kind of like a first crack at how we can kind of align to different distributions of data right um they they're only going to affect like pre and post processing and they're really only targeting like let's say specific statistics right like mean and and variance and so on um and so just because you align the mean and the variance of two data sets it doesn't mean that the two data sets are the same right they have the same distribution right there could be other like higher order factors right that are not taken into account right um and so the the kind of the Crux of the idea behind our method um Ben or batch effects mization is that we are going to apply uh normalization at many different layers of a neural net um so uh the the nice thing about this is that it's simple and it requires like minimal changes to existing pipelines um but the the idea being that when you apply this to multiple layers of a neuron net you're kind of giving like different transformations of the input data an opportunity to correct for those batch effects right and so if you correct at like so here what I've done is I've highlighted all layers in which we're kind of applying standardization we're exploiting like a commonly found layer in neural Nets called match Norm um and basically by applying it at multiple layers of a neural Nets you're giving the the data at various various scales right a chance to standardize the data hopefully align those batch uh statistics in a in a better manner um so maybe I can give a quick primer on like bat Norm so bash Norm is like a

layer that we commonly find already in a lot of Deep learning models and so we're we're going to repurpose this layer for batch effects correction um so so one point important point that I want to make is that batch Norm um was originally invented to stabilize training it had nothing to do with correction for batch effects right it's something that kind of rose out deep Learning Community they weren't really thinking about like biological data with batch effects um so in a in a typical setup of how it works so let's say you have a data data set here of like microscopy images um what you would do is you know you You' randomly sample a training batch so this training batch it's it's like a different batch than like experimental batch it's like an overloaded term that deep learning people also use um it basically just means this is a set of Sub set of data that I'm going to feed to a neural network at a particular training iteration um and so what happens is I I take this data I feed it to my M and then within each batch Norm layer So within like the red boxes that I've highlighted there what ends up happening is basically standardization right so you take your data points and you standardize by mean and um mean and also the standard deviation then then you get an output right and so the like I said the original motivation for this was to equalize the the features the different features along the dimensions it had nothing to do with batch effect correction um what we notice is like there is kind of one simple tweak you can make to this picture so to repurpose these batch Norm layers for batch effects correction and that's that's simply that when you train your neural Nets instead of sampling randomly what if you sampled like exclusively from one experimental batch so you align like what it means for a training batch like in deep learning with a experimental batch in biology right so then what ends up happening is that when you apply the standardization these batch norm layers right your means of standard deviations they kind of absorb uh statistics having to do with your experimental batch and then um you can kind of standardize out those statistics when you do the the normalization um so this picture kind of paints Rosy picture right like like like you you probably won't eliminate all batch Effects by just applying this normalization once right that's why classical standardization techniques don't work but the fact that you do this at every layer kind of makes makes it so that you're standardizing over and over and over again and giving your network a chance to correct at many different layers um so that's that's like the the basic per of the idea and so um oh and this is just basically showing that at the next iteration you would sample exclusively from a different batch so like let's say the blue batch here and so you would be able you would be able to kind of standardize out the effects of the blue batch so you do this during training and and so this is the Crux of the idea and so there's really only like three like simple steps involved um one is that uh during training which I've already shown you you sample exclusively from experimental batch when constructing your deep learning batch um and then secondly during inference so when you test the model um so what is typical you do what people typically do is they feed like one input in at a time right um so we say don't do that instead you want to do basically what what you did during training right you want to take an entire experimental batch and feed it into the model and this will allow you to use the the batch Norm to kind of do exactly what you did during training which is you standardized by the batch statistics as opposed to like some Global statistics that summarize whole data set um so this way you can kind of remove um remove batch effects from U new batches during

train uh during inference as well so that that's that's like the core idea and then I just wanted to share some results so we basically tried to apply this method um we're try to valate this method in very in many different settings so in supervised learning and unsupervised learning and transfer learning settings um and trying to see like does this actually improve performance of of neural networks trained for microscopy um so the the data set we worked with is called rxr exing Wilds um so it's developed by recursion and Wilds is like a subsampling recursion just to make it more comp computationally tractable um but overall there's like 125,000 microscopy images collected over 200 and for like different plates you can think of each plate as a batch that's that's what we treated um and each image right so there an example here it contains like cells from a particular cell line exposed to one of 10,000 or so genetic treatments sna treatments so these are treatments that um target mRNA to try to degrade it after uh transcriptions to try to prevent translation um and uh and then the the task here that that the data set proposes is to take the image take images and try to predict like the there are a few things that make this Tas difficult so the first is that there's a large number of classes right there's thousand or so classes so if you were to randomly like choose a class you wouldn't You' get like less than 1% accuracy um and then secondly uh there are different cell lines right so the model has to be able to integrate across different cell lines perform well and then uh finally like the the way that they split their training and evaluation is that you have to train on different plates that you use for evaluation right so that's they're entire new batches of data that you don't see so you can't just latch on to batch effects to perform well and in fact to do well on this task you have to be able to overcome batch you have to be able to generalize new batches um and uh yeah so so in the symol classification task um basically what we have what we showed so if you train a standard like resnet on this data set you get like 30% accuracy um so with just applying like our technique um you get a bump by 10% to 40% accuracy and this is like on par with um some of the best reported numbers for this data set um and one thing I want to highlight is that um what we like about our method is that it's very simple right so it's it's really just about rearranging the way that you kind of assemble training batches during um model training um so there's no need to like for example create a s Network or uh to heavily use dat data augmentation to kind of corre for batch effects um and then we did a few uh uh abolition studies to kind of just show that like um certain factors are are important for maintaining the performance so so the first is showing that you know if you don't use like a large enough batch right if you don't take a large enough like like batch for your experimental batch during um training or during evaluation right um you get like noisy estimates of your batch statistics so then your performance drops um so it's important to have like a large enough batch to feed into your model during valuation the second thing is also to show that it's important to correctly Define what is a batch um so you can imagine like okay I have like plates of images but maybe I also collected certain plates on like the same day right so I maybe I I assume like maybe all the plates I collected on the same day are like one batch right um but we we saw that you know if you group like your your batches in a different way um you could get like a very different levels of performance so it's important to like properly Define like what is the kind of overarching batch effect is it plate wise or is

it's like kind of experiment level um and then we also so that was supervised learning but then we also tested this on like a unsupervised learn algorithm so I mentioned for morphologic morphological profiling there's like a whole slew of algorithms some of which are developed here at the Bro right for how do you take images and create these feature vectors so we we evaluated one very popular one which is called uh Sim clear um so here's a schematic of basically what SIM clear is and the the specific detail aren't that important but basically the high level idea is that it learns representations that are invariant to like you augmenting the image so like rotating it or cropping it and so on right so it's you can think of it's just so so there's some algorithm that kind of gives you um this feature extractor um and uh basically we wanted to see like what was the net effect of using like our batch effects correction method then during training of sim clear right um and um so so what we did was you know we ran Sim clear we we got the feature vectors and then we try to see how predicted those are of treatment remember the features were not trained with any knowledge of treatment labels because it's unsupervised so so we want to see like how predictive are they of treatment um using a linear classifier and then we we compare that to what you did if you trained it with um our batch effects correction so um what we saw is that if you just use sim clear you get like 11% accuracy so these these numbers are going to be lower than the ones I showed on previous slide because this is unsupervised right there's no knowledge of treatments that were used to create the features um but then by using then you can kind of um jump up to like 19% accuracy and then we also uh kind of repeated this with a newly developed unsupervised learning algorithm that we thought would be better than Sim clear specifically adapted to microscopy images we call it multi instance contactive learning or Min clear again the details are not super important um but basically it's like a different if think of it as a different um unsupervised learning algorithm that is specifically adapted to this setting of mosop images um we showed that you know M CLE has some baseline performance of like 13% and then when you apply Bend to it you again like close to double its accuracy um so this was like all encouraging that like you know we're doing something good it seems like with our batch effects correction method um but let's like what we did also was we did a further analysis we took a closer look at kind of these feature SPAC right and what we were learning um and um basically looking at like those feature Spaces Project projected in two Dimensions we saw that like the standard versions of these methods you you can see that the batch effects are very prominent um because you have like clusters of images based on the batches but then um with then you're able to kind of smooth some of these out um and uh yeah you see a more evenly dispersion among the different backs um and then um one one other thing I want to mention is that uh so with batch effects correction methods uh one common um I guess one one common thing that we worry about is that maybe they correct for too much sometimes right right like if they you know you can correct for batch effects but if you eliminate meaningful biological signal then that's also bad um so here at least we have like a small example of how even though we're corre for batch effects we are preserving some um useful biological information um in this case it's like cell lines so I mentioned how the the data set had different cell lines right um so what we saw is if you look at like the with Ben plot right it would um the the representations they would be merged together

for batches of the same cell line but for different cell cell lines like those distances can still be preserved so there's still some like biological information that's preserved um and I think it's it's important to like make sure we validate this way to make sure that we don't um you know kill off all useful signal when we use batch effects correction right I was talking to shanon thing yesterday who was saying like yeah the best batch Direction method is one that kind of replaces everything with all zeros right because that then you kind of equalize everything right but but that also kills a lot of um cool so and then finally we also EV validated this on transfer learning so this is something very similar to like kind of the setup that stanless describing earlier um where we train on one data set and we try to evaluate on different data sets um and um so we trained on on RX RX1 the recurring data set and we evaluate on data set is developed here at the broad c021 meis of action um classification and um we see kind of similar Trends so um you know with with kind of your standard uh transfer learning uh setup get a certain percentage then you get a boost um by using our batch effects correction method um and this is like comparable to some of the um some of the better methods that we or better numbers that we see from like other data sets transferring in um and then a final thing that I thought was interesting is that so for this data set there's like a well-established batch effects correction method which is typical variational lation that um a lot of people use and it's like these methods other methods here they also use and so um just kind of exploring right what is the effect of typical variation normalization on our method um so we saw that you know comparing the red to the red right if you apply typical variation normalization to like a method uh to a model train without Bend then you get a slight boost in performance um but when you apply it to a model with Ben you see no change in performance um and then we also have other settings where we validate kind of the the same thing and so overall this trend seem to suggest that Ben is kind of subsuming some of these um uh techniques that have been used in typical variation realization like it's kind of doing um kind of similar things um because like when you apply them on top it doesn't seem like there's there's any added benefits and that kind of makes sense because both of these like are kind of rooted in standardization um cool and let me just summarize real quick so we introduced batch ex realization as a method to help learning models correct for batch effects um so uh we're going to highlight that it's it's very simple it's very easy to implement it's just about how you rearrange data when you train these models um and we show that it's applicable to several types of of deep learning and that it led to some performance gains on um mic processing data sets in terms of future work and limitations I think there are are many inter Avenues potentially so um I really like this figure here this is by Juan Caso here at the rooad um basically showing that you have you know batch effects are not just limited to like you know this plate has one effect another plate has another effect right like there are more like Nuance effects so if you look for example in play two you'll notice that on the edges there are some batch effects right the Wells on the edges have certain batch effects different from like the ones in the center um so this is something that our method currently like cannot deal with right right so like how do you correct for you know some of these more Nuance batch effects that occur for example within a plates um uh there's also like computational I guess uh challenges that that that I think would need to

be addressed so one is that like I I showed you a dation study earlier where uh the method kind of performs worse if you don't have like a full a large like batch size at at imprint time right to be able to calculate those test statistics um so that that definitely requires a lot of to be able to feed that into the model at once so uh maybe there there are ways to kind of reduce that and then from a statistical pers perspective like uh there will be times when we collect batches of data that just don't have a lot of data points so again like there will be some um maybe estimation error in like those batch statistics that we're using at every layer of our newal net standardized data so you know how can we like still assure good performance even when that happens I think that's also question